Mehra and Prescott's original skepticism was not without merit. Looking only at U.S. data, there has not been a large drop in consumption, except for the Great Depression in the 1930s, and even then it was not catastrophic.[6] Looking more broadly at data from other countries and using a long time series, however, Barro and Ursua (2011) document some sizeable drops in consumption and GDP—even for countries that are large and developed today. For example, Barro and Ursua report that Germany's consumption declined by 41% in 1945 and Japan's fell by 50% during World War II. The United States and the United Kingdom have been more tranquil, with the largest decline being 16% for the United States in 1921 and 17% for the United Kingdom in 1945. But these are exceptions. For many countries, macroeconomic disasters have been truly calamitous. Russia's consumption fell 71% in World War I and the Russian Revolution and again by 58% during World War II (poor Russia!). China's GDP (not consumption) fell 50% from 1936 to 1946, and Turkey's consumption fell 49% during World War II.

The disaster explanation is also labeled a *peso problem*, which refers to a very low probability event that is not observed in the sample. The U.S. equity premium is high because it reflects the probability of a disaster, but so far we have not observed that disaster. The first published use of the term "peso problem" appears in Krasker (1980) in the context of the phenomenon that the Mexican peso traded at a steep discount on the forward market during the early 1970s. It appeared to be mispriced, but market participants were expecting in a devaluation that eventually occurred in August 1976. Researchers only looking at the early 1970s sample would have concluded that an asset price—the forward peso—appeared anomalous because a crash had not occurred in the sample available to them. Likewise, the U.S. equity premium may appear too high because we have not observed a disaster in the U.S. sample.

If equities command a risk premium because of infrequent crash risk, then the pertinent question to ask an investor is how he can weather such disasters. Of course, there are some disasters so ruinous that an entire society disappears, and a new one takes its place, such as the Russian Revolution in 1918 and the rise of modern Communist China, where all domestic investors are wiped out. (This by itself should be a good reason for diversifying across countries as mean-variance investing advocates; see chapter 3.) The fact that we have not seen a truly terrible consumption disaster in the United States should give us pause. The extreme advice implied by this theory is that equity investors should stockpile AK-47s and MREs in their basements in case society collapses. Before the cataclysm comes, equity returns will be high; that equity returns are in fact high, on average, foretells the cataclysm to come. We just haven't seen it yet.

[6] For the entire economy. Certainly it was catastrophic for certain individuals like the Joad family, in John Steinbeck's *Grapes of Wrath*.